

From Latent Trajectories to Closed State Channels: What Long-Horizon Reasoning Requires Beyond One-Step Competence

Anonymous authors

Abstract

Language models often solve individual reasoning steps but fail when the same steps must compose. We ask whether this gap reflects a missing computational interface: the model may represent task state without making it sufficient, shared across contexts, and closed under repeated updates. On balanced synthetic state machines, Qwen2.5-32B reads and updates three-bit states perfectly, yet scores 13.44% on a two-step history followed by a final lookup. Deleting the history and passing only the model’s predicted state raises accuracy to 76.98%. We then train Qwen2.5-7B through one-token state channels with known information rates. A two-bit channel reaches its exact 50% recovery ceiling. With matched compute, training the state transition itself rather than history endpoints raises a redundant four-bit channel from 52.50% to 97.08% at horizon 16 and retains 93.25% mean accuracy across five shifted history families. Harder register programs expose the limit: about 89% conditional one-step reliability compounds to 7.92% final accuracy at horizon 32. A small proof-state pilot suggests redundant codes can help at greater active deduction depth, but needs replication. These results separate state information from a usable state interface and connect latent reasoning trajectories to rate, closure, and reliability.

1 Introduction

Work on reasoning trajectories asks what changes inside a model as it moves through a solution (Sun et al., 2026). Work on implicit discrete state representations asks whether hidden activations track the intermediate values needed for symbolic computation (Chen et al., 2024). Both views leave a further question: can the representation serve as a computational interface?

We distinguish four properties. A state code must contain enough information about the task state; it must be sufficient for future outputs once the past is removed; independent producers and consumers must attach the same meaning to it; and applying a legal update to a model-produced code must yield another valid code with the right meaning. We call the last property *closure*.

Our central result is not that a token bottleneck alone improves reasoning. It is that rate and closure predict whether a learned state channel survives recursive use. The result is strongest for finite addition state machines and fails on mixed 16-state register programs. The failure is informative: high one-step accuracy is not equivalent to a reliable long-horizon module.

Contributions.

1. We give a behavioral diagnosis in which local state operations are perfect but native serial composition fails.
2. We define an explicit, one-token intervention that removes the original history and tests whether a state alone controls the continuation.
3. We vary channel rate, test code interchange, and compare transition closure against a token- and forward-matched endpoint control.
4. We measure how local error compounds on addition, register, and proof state machines, and state the boundary between the confirmed result and the present pilots.

2 State interfaces in reasoning trajectories

A history $H = (u_1, \dots, u_T)$ acts on a finite state $S_t \in \mathcal{S}$, and a final rule F maps S_T to answer Y . The model emits a one-token code $C \in \mathcal{C}$. A decoder $q : \mathcal{C} \rightarrow \mathcal{S}$ defines the code’s semantic quotient. We test:

$$Y \perp H \mid (C, F) \quad (\text{sufficiency}), \quad (1)$$

$$q(\widehat{T}_u(c)) = T_u(q(c)) \quad (\text{closure}), \quad (2)$$

$$R = \log_2 |\mathcal{C}| \quad (\text{declared rate}). \quad (3)$$

This is a finite predictive sufficient statistic (Shalizi and Crutchfield, 2001). It also gives a controlled information bottleneck (Shannon, 1948; Tishby et al., 2000): unlike a hidden vector, the channel has a declared maximum capacity.

Why this differs from probing. A probe asks whether state can be decoded from an activation. Our intervention deletes the history, passes the code to a separate call, and checks whether changing the code changes the answer as the substituted state requires. Information can be decodable without satisfying this causal contract.

3 Tasks, prompts, and controls

3.1 Three-bit addition and pointer lookup

The state is an integer in $\{0, \dots, 7\}$. History operations add modulo eight. A final pointer table permutes the eight answers. For example:

Start at 2; add 0; add 6; FINAL is [3,5,7,6,2,0,4,1]. The history endpoint is 0 and the answer is 3.

Read, Update, Synthesize, and Compose use separate prompts and one exact output token. Explicit handoff uses two calls: $H \rightarrow C$, then $(C, F) \rightarrow Y$. The second call contains no history. Recursive evaluation uses $(C, u_t) \rightarrow C'$ and discards the used operation.

3.2 Training

Rank-16 LoRA adapters update attention and MLP projections. Prompt tokens are loss-masked. Interface training has one producer and one consumer forward per semantic program, each with one supervised target token. Matched controls use the same programs, forward count, target count, fixed padding, and optimizer. Program contexts for producers, consumers, and tests are disjoint.

3.3 Reasoning state machines

Register programs use two two-bit registers and add, XOR, swap, and conditional instructions. The final rule is an unseen 16-way dispatch table. Proof programs use four fact bits as a proof frontier; rules update the frontier and a binary query reads it. In the active-depth probe, all programs contain 64 rules while the number of rules that add a fact varies from zero to four.

4 Results

4.1 Native trajectories contain operations without serial routing

Figure 1 shows the main diagnosis. At h2, Synthesize reaches 76.98% while Compose reaches 13.44%. Among cases where Read, Update, and Synthesize all succeed, Compose remains 13.59%. This is a routing failure, not an absence of the local operation.

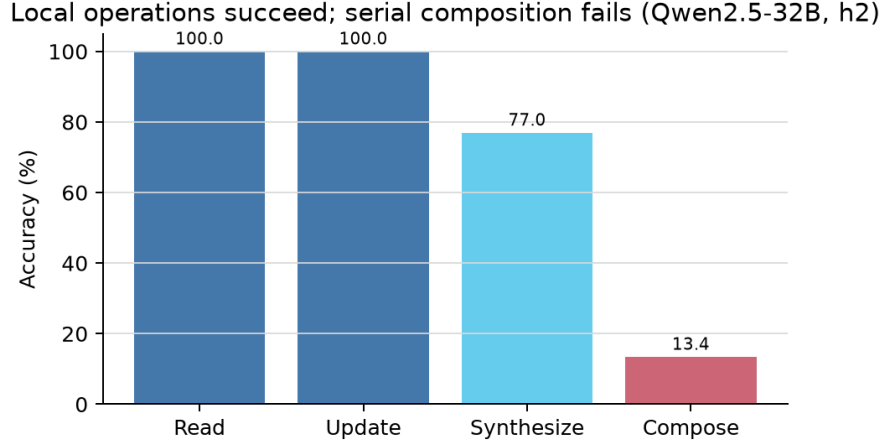


Figure 1: Frozen Qwen2.5-32B at horizon two. All 1,920 cases pass the local competence controls, yet Compose remains near eight-state chance.

4.2 Explicit state handoff repairs the immediate failure

At h2, model self-handoff matches state synthesis at 76.98%, a +63.54 point gain over one-pass Compose. Gold handoff and stepwise decimal execution are perfect. At h4, state synthesis itself falls to chance, so self-handoff cannot repair the upstream error.

4.3 Rate limits recovery and redundant rate improves reliability

The two-bit channel maps pairs of states to the same code and reaches exactly 50% answer accuracy at every horizon. The context-bound control remains near 12.5%, despite having enough surface symbols, showing that a shared dictionary matters. Gold-code swaps preserve same-state behavior and transfer different-state behavior for the canonical and redundant codes.

4.4 Closure, not endpoint fit, drives long-horizon reuse

At h8 and h16, redundant closure beats endpoint training by 43.85 and 44.58 points, with program-context bootstrap intervals excluding zero. Under five history shifts, closure retains 93.25% mean h16 accuracy versus 48.50% for the endpoint control. This rules out a single positional history template, but not overfitting to the finite addition transition family.

4.5 Local reliability is not global reliability

The register confirmation fails its preset gate. Conditional transition accuracy is 89.31%, but complete interface accuracy is 7.92%, against 5.83% one-pass and 6.25% chance. The programs contain about 24 real state changes. An extra channel bit raises two completed seeds to only 12.81% and 12.19%. Redundancy cannot compensate for a transition kernel that is not reliable enough under mixed recursive use.

4.6 Proof-state pilot

The redundant five-bit proof interface is 81.67% overall and reaches 100% at four active deductions, versus 75% one-pass. Canonical four-bit accuracy falls to 33.33% in the same stratum. This is consistent with redundancy improving recursive reliability, but the small, non-monotone strata do not yet establish a rate-reliability law for proof.

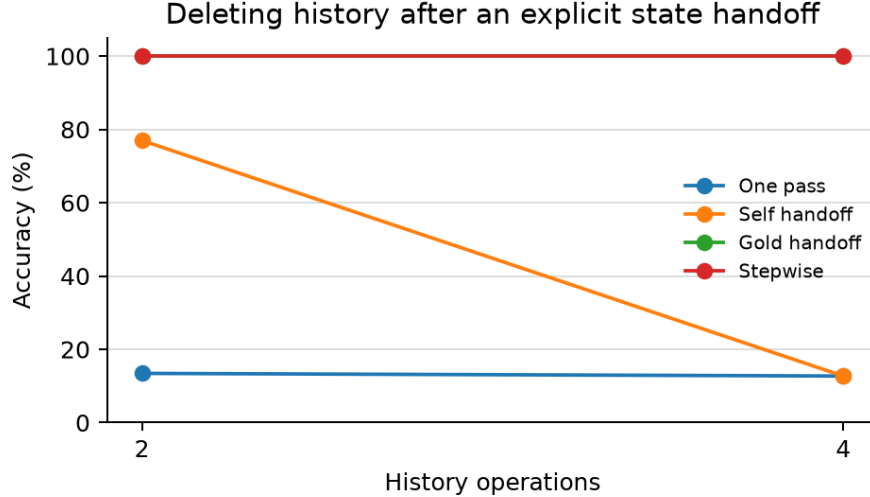


Figure 2: A second call receives only the state and final rule. Gold and stepwise conditions show that state use and local updates are available.

5 Relation to latent state and long-horizon reasoning

Chen et al. (2024) show that models can form implicit discrete state representations and use them in arithmetic. We test a different boundary: whether state information is organized as a reusable interface. Our frozen model result shows that local decoding and local manipulation do not imply serial routing. Our trained result shows that an explicit channel can acquire sufficiency and closure. The register failure shows why long-horizon reasoning needs more than state identification: if each self-conditioned update succeeds with probability below one, error compounds with effective transition depth.

This suggests that reasoning trajectories should be studied at three levels: representation geometry, semantic state, and interface dynamics. Geometry asks where information appears. State asks which histories are equivalent for the future. Interface dynamics asks whether the model can repeatedly move between those equivalence classes without leaving them.

6 Limitations and falsifiers

The strongest result uses finite synthetic addition. The proof result has 12 cases per depth and one model checkpoint. The register result is a negative confirmation, not a repair. We do not claim that ordinary language models naturally expose discrete state tokens, that extra capacity always helps, or that the method yet improves open-domain reasoning.

The main claim would weaken if matched closure controls reproduced the gain, if independently trained consumers could not use the codes, if code swaps did not transfer state-dependent behavior, or if the rate ceiling failed under balanced states. Broad reasoning claims require tasks where the sufficient state is not supplied by the benchmark designer.

7 Next experiments

1. Confirm the proof-depth comparison over at least 30 contexts and three seeds, balancing target states within each active-depth stratum.
2. Train mixed-operation register transitions to a preset local reliability target before evaluating horizon, then compare observed end accuracy with a measured error-propagation model.

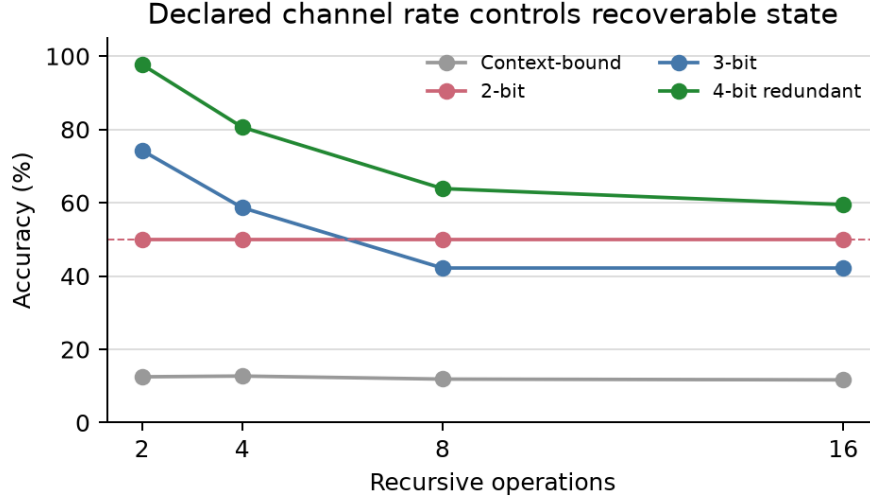


Figure 3: One-token channels with declared rates of two, three, and four bits for an eight-state task. The two-bit channel reaches its exact 50% ceiling.

3. Test proof search, program execution, and planning tasks with explicit sufficient-state definitions and held-out transition compositions.
4. Connect token interfaces to latent trajectories by measuring when a hidden trajectory becomes predictive of the emitted quotient state and whether interventions at that point change the code.

8 Reproducibility

The paper workspace’s `evidence.json` maps every plotted number to its saved JSON source. Complete and partial runs remain separate. Data generation uses fixed seeds, hashed program files, group-disjoint contexts, exact symbolic verification, and program-context bootstrap intervals. No LLM judge scores these tasks.

References

- Junhao Chen, Shengding Hu, Zhiyuan Liu, and Maosong Sun. States hidden in hidden states: Llms emerge discrete state representations implicitly. *arXiv preprint arXiv:2407.11421*, 2024.
- Cosma Rohilla Shalizi and James P. Crutchfield. Computational mechanics: Pattern and prediction, structure and simplicity. *Journal of Statistical Physics*, 104:817–879, 2001.
- Claude E. Shannon. A mathematical theory of communication. *Bell System Technical Journal*, 27:379–423, 1948.
- Yuxuan Sun et al. Llm reasoning as trajectories: Step-specific representation geometry and correctness signals. *Manuscript*, 2026.
- Naftali Tishby, Fernando Pereira, and William Bialek. The information bottleneck method. In *Allerton Conference on Communication, Control, and Computing*, 2000.

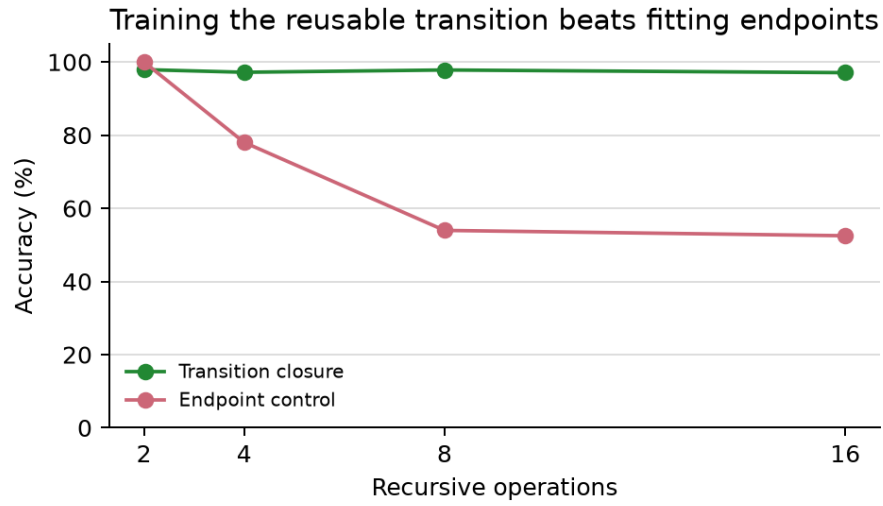


Figure 4: Matched training compute; only the producer target differs.

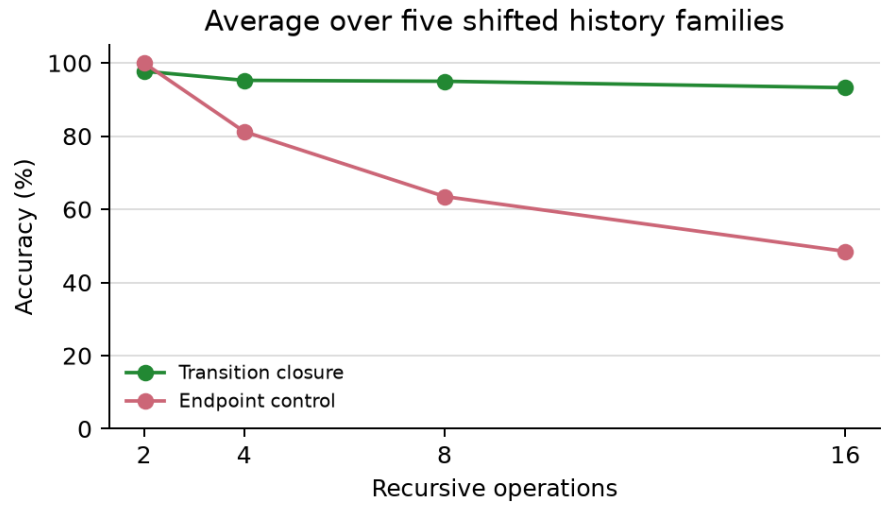


Figure 5: Mean accuracy across structured, IID, shuffled, cancellation, and repeated-operation histories. Each cell has five program contexts.

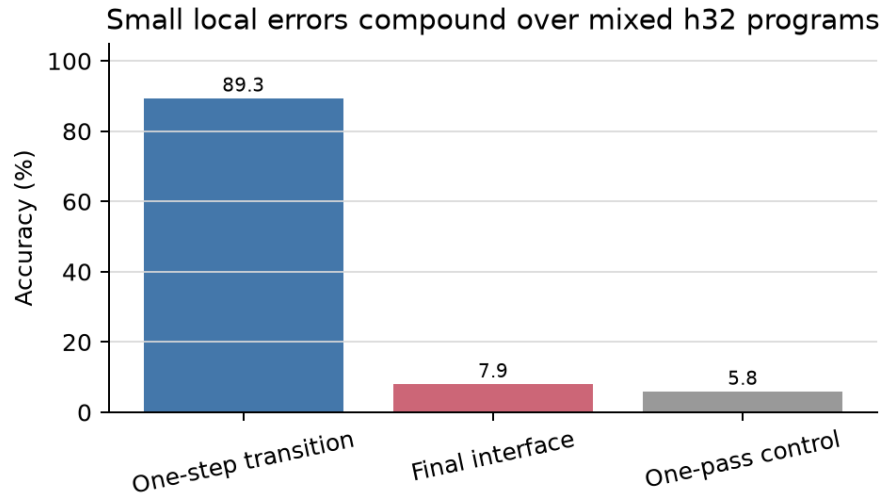


Figure 6: Three Qwen seeds on held-out mixed register programs at h32. The one-step score conditions on the state code the model actually received.

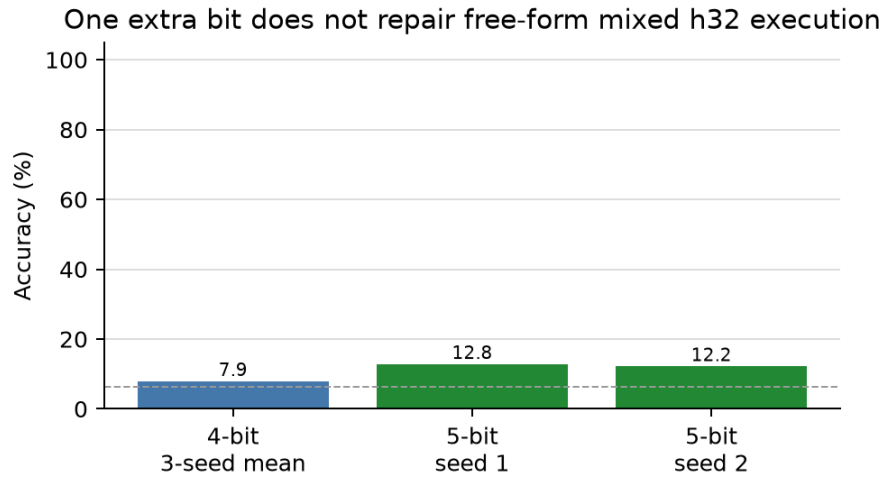


Figure 7: Held-out mixed h32 register accuracy. The third five-bit seed has no complete summary and is excluded. The dashed line is 16-state chance.

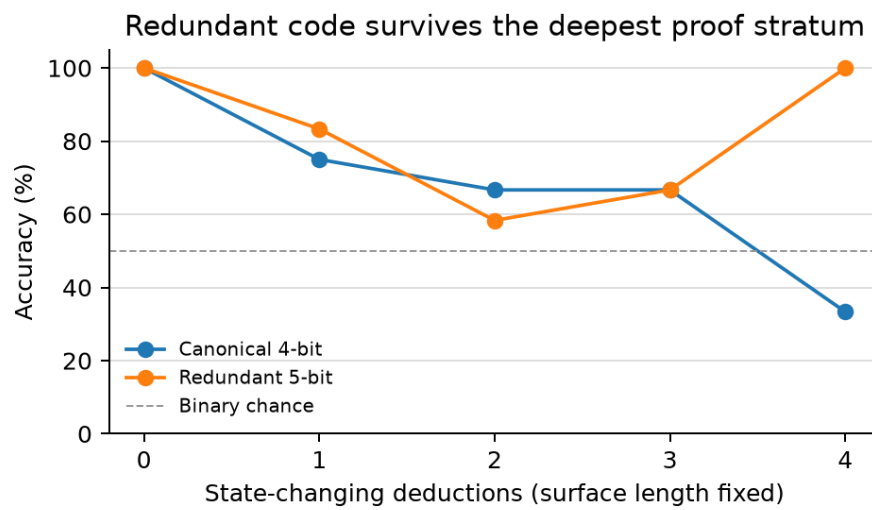


Figure 8: Fixed 64-rule surface length; each point has 12 cases.